A Major Project Report on

Efficient In-Line Data Deduplication for Non-Volatile Memory Storage Using I/O Causality

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Jawaharlal Nehru Technological University, Hyderabad

in partial fulfillment of the requirements for the award of the degree

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING

by

B VENKATA ABHINAV	**21831A0530**
D SAI KALPANA	**21831A0545**
D SRILATHA	**21831A0547**

Under the Esteemed Guidance of

Mrs. P. Anusha

Assistant Professor, CSE

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

GURU NANAK INSTITUTE OF TECHNOLOGY
(Affiliated to JNTUH-Hyderabad)
Ranga Reddy District-501506

2024-2025



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Ranga Reddy District-501506

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CERTIFICATE

This is to certify that the project entitled **Efficient In-Line Data Deduplication for Non-Volatile Memory Storage Using I/O Causality** is being presented with a report by **Buragadda Venkata Abhinav (21831A0530), Dodda Sai Kalpana (21831A0545), Dummadi Sri Latha (21831A0547)** in partial fulfillment for the award of **Degree of Bachelor of Technology in Computer Science and Engineering,** to **Jawaharlal Nehru Technological University, Hyderabad**.

Mrs. P. ANUSHA
INTERNAL GUIDE

Dr. B. SANTHOSH KUMAR
HEAD OF DEPARTMENT

EXTERNAL EXAMINER

RAM Innovative Infotech
M : +91 9581 012 012
E : raminnovativeinfotech@gmail.com
Flat No.#309, Amrutha Ville,
Opp: Yashoda Hospital, Somajiguda,
Hyderabad-82, Telangana, India
Www.raminnovativeinfotech.webs.com

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PROJECT COMPLETION CERTIFICATE

This is to certify that the following students of final year B. Tech, Department of Computer Science and Engineering - Guru Nanak Institute of Technology (GNIT) have completed their training and project at GNIT successfully.

	STUDENT NAME:	Roll No:
1.	Buragadda Venkata Abhinav	21831A0530
2.	Dodda Sai Kalpana	21831A0545
3.	Dummadi Srilatha	21831A0547

The training was conducted on Java Full Stack Technology for the completion of the project titled Efficient In-Line Data Deduplication for Non--Volatile Memory Storage Using I/O Causality in 2024-2025. The project has been completed in all aspects.

RAM INNOVATIVE INFOTECH HYDERABAD

Signature

DECLARATION BY STUDENTS

We hereby declare that the major project report entitled **"Efficient In-Line Data Deduplication for Non-Volatile Memory Storage Using I/O Causality"** is the work done by **Buragadda Venkata Abhinav, Dodda Sai Kalpana, Dummadi Srilatha** bearing the roll no's **21831A0530, 21831A0545, 21831A0547** towards the fulfilment of the requirement for the award of the **Degree of Bachelor of Technology in Computer Science and Engineering**, to **Jawaharlal Nehru Technological University, Hyderabad,** is the result of the work carried out under the guidance **Mrs. P. Anusha**, Assistant Professor of Computer Science and Engineering **Guru Nanak Institute of Technology, Hyderabad.**

I confirm the following:

1. I have contributed the entire project process, from the selection of the project title to the submission of the final report.
2. The project report has been prepared as per guidelines, ensuring adherence to high standards of quality, clarity, and structure.
3. All information provided and work done is completely genuine and is not copied from any online or offline platform.
4. That I or any of my known have not submitted this project to any other University or College.

I further certify that this project report has not been previously submitted in part or full for the award of any degree or diploma by any institution or university.

21831A0530	21831A0545	21831A0547
B.VENKATA ABHINAV	**D.SAI KALPANA**	**D.SRILATHA**

DECLARATION BY GUIDE

I, Mrs. P. Anusha , hereby declare that I have guided the major project titled " Efficient In-Line Data Deduplication for Non-Volatile Memory Storage Using I/O Causality " Undertaken By Buragadda Venkata Abhinav(21831A0530), Dodda Sai Kalpana(21831A0545),Dummadi Srilatha(21831A0547). This project was carried out towards the fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering, to Jawaharlal Nehru Technological University, Hyderabad.

As the guide, I confirm the following:

1. I have overseen the entire project process, from the selection of the project title to the submission of the final report.
2. I have reviewed and certified the accuracy and relevance of the results presented in the report.
3. The work carried out is original, free from plagiarism, and adheres to ethical guidelines.
4. The contributions of each student have been appropriately recognized and assessed.
5. The project report has been prepared under my supervision, ensuring adherence to high standards of quality, clarity, and structure.

I further certify that this project report has not been previously submitted in part or full for the award of any degree or diploma by any institution or university.

Name of Guide : **Mrs. P. Anusha** Signature of the Guide
Date:
Place:

Name of HOD: **Dr. B. Santhosh Kumar** Signature of the HOD
Department: **Computer Science & Engineering**
Date:
Place:

A64CKNOWLEDGEMENT

"Task successful" makes everyone happy. But happiness will be gold without glitter if we don't state the people who have supported us to make it a success.

We would like to express our sincere thanks and gratitude to **Dr. B. SANTHOSH KUMAR**,professor & Head - Department of Computer Science and Engineering, Guru Nanak Institute of Technology for having guided us in developing the requisite capabilities for taking up this project.

We thank the Project Coordinator **Mr. D. SRINIVAS,Assistant Professor** Computer Science Engineering, Guru Nanak Institute of Technology for providing seamless support and the right suggestions that are given in the development of the project.

We especially thank our internal guide **Ms. P. ANUSHA**, Assistant Professor for his constant guidance in every stage of the project. We would also like to thank all our lecturers for helping us in every possible way whenever the need arose.

On a more personal note, we thank our beloved parents and friends for their moral support during our project.

CO-PO MAPPING

COs	Course Outcome	BL
CO1	Independently carry out literature survey in identified domain, and consolidate it to formulate a problem statement	4
CO2	Apply identified knowledge to solve a complex engineering problem and design a solution, implement and test the proposed solution	3
CO3	Use synthesis/Modeling to simulate and solve a problem or apply appropriate method of analysis to draw valid conclusions and present, demonstrate, execute final version of project	3
CO4	Contribute effectively as a team member or leader to manage the project timeline and Incorporate the social, environmental and ethical issues effectively into solution of an engineering problem	6
CO5	Write pertinent project reports and make effective Project Presentations	3

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO 1	–	3	–	–	–	–	–	–	–	–	–	3	3	3	3
CO 2	3	–	3	–	–	–	–	–	–	–	–	–	3	3	3
CO 3	–	–	–	3	3	-	-	-	–	–	–	–	3	3	3
CO 4	–	–	–	–	–	3	3	3	3	–	3	–	–	3	–
CO 5	–	–	–	–	–	–	–	–	–	3	–	–	–	3	–



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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

VISION

To be recognized as a leading department of Computer Science and Engineering in the region by students, employers and be known for leadership, Ethics, and commitment to fostering quality teaching-learning, research, and innovation.

MISSION

- Nurture young individuals into knowledgeable, skill-full and ethical professionals in their pursuit of Computer Science and Engineering.
- Nurture the faculty to expose them to world-class infrastructure.
- Sustain high performance by excellence in teaching, research and innovations.
- Extensive partnerships and collaborations with foreign universities for technology upgradation.
- Develop Industry-Interaction for innovation and product development.

ABSTRACT

I/O causality-based in-line data deduplication (ICID) is a novel approach designed to enhance deduplication efficiency in non-volatile memory (NVM)-enabled storage systems. Traditional chunk-based offline deduplication methods suffer from significant performance overhead due to data chunking and indexing, making them unsuitable for NVM-based storage, which benefits from fine-grained, byte-addressable operations. Unlike conventional in-line deduplication schemes that rely on hash indexes, ICID leverages a B-tree structure to record memory copy operations, enabling causality-based deduplication. To optimize efficiency, ICID introduces two key techniques: grouping memory-copy records targeted to the same page within a B-tree node to enhance data locality and exploiting spatial locality to identify and remove outdated memory-copy records, thereby reducing memory consumption. Evaluations conducted on a system with Intel Optane DC Persistent Memory Modules demonstrate that ICID significantly improves deduplication performance, achieving up to 16× higher deduplication ratios and reducing deduplication time by an average of 47% compared to state-of-the-art methods, particularly benefiting key-value stores like LevelDB.

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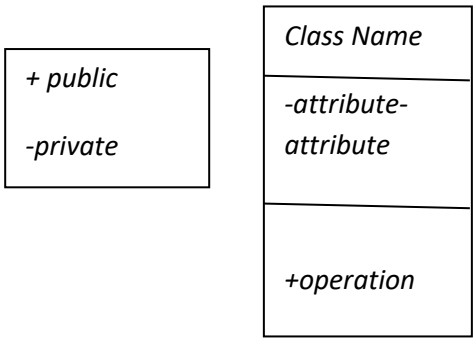
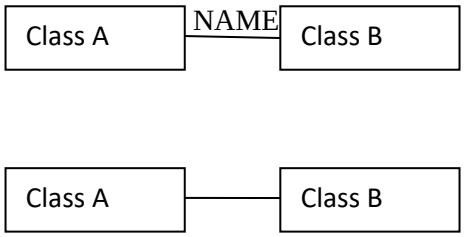
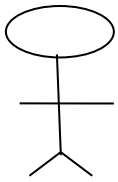
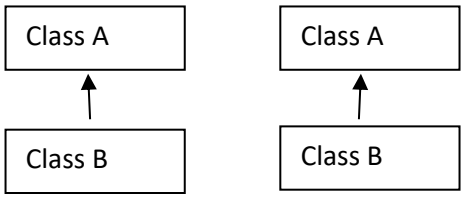
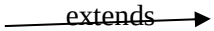
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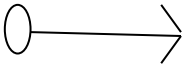
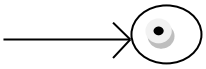
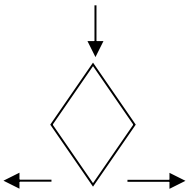
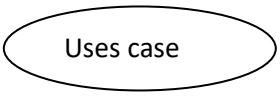
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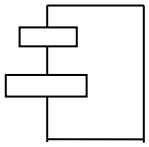
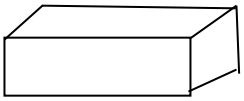
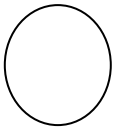
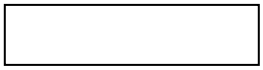
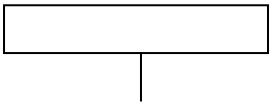
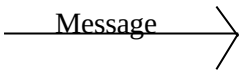
LIST OF ABBREVATIONS

S.NO	ABBREVATION	EXPANSION
1.	DB	Data Base
2.	JVM	Java Virtual Machine
3.	JSP	Java Server Page
4.	CB	Collective Behavior
5.	RSSS	Ramp secret sharing scheme
6.	JRE	Java Runtime Environment

LIST OF NOTATIONS

S.NO	NAME	NOTATION	DESCRIPTION
1.	Class		Represents a collection of similar entities grouped together.
2.	Association		Associations represents static relationships between classes. Roles represents the way the two classes see each other.
3.	Actor		It aggregates several classes into a single classes
5.	Aggregation		Interaction between the system and external environment
6.	Relation		Extends relationship is used when one use case is similar

	(extends)		to another use case but does a bit more.
7.	Communication		Communication between various use cases.
8.	State		State of the process.
9.	Initial State		Initial state of the object
10.	Final state		Final state of the object
11.	Control flow		Represents various control flow between the states.
12.	Decision box		Represents decision making process from a constraint
13.	Usecase		Interact ion between the system and external environment.

14.	Component		Represents physical modules which is a collection of components.
15.	Node		Represents physical modules which are a collection of components.
16.	Data Process/State		A circle in DFD represents a state or process which has been triggered due to some event or action.
17.	External entity		Represents external entities such as keyboard, sensors, etc.
18.	Transition		Represents communication that occurs between processes.
19.	Object Lifeline		Represents the vertical dimensions that the object communications.
20.	Message		Represents the message exchanged.

CHAPTER 1

INTRODUCTION

1.1 GENERAL

DATA deduplication techniques have long been studied for space saving in storage systems. They eliminate duplicate data at file or chunk levels by identifying the same data using cryptographic hash functions such as SHA-1 or MD5. Data deduplication can be performed in an offline or inline mode. The offline deduplication is conducted after the data has been written to the storage device, and thus usually causes write amplification. In contrast, the inline deduplication detects the duplicate data before writing it back into storage, and thus can reduce data redundancy and the wear of storage devices. More importantly, inline deduplication has the potential to explore the correlation among I/O operations at runtime, and may significantly improve the deduplication ratio. Non-volatile memory (NVM) technologies such as Intel 3D Xpoint offer much lower cost, higher density and energy efficiency than traditional DRAM technologies. They have become a promising complement to DRAM to bridge the performance gap between main memory and SSD/HDD storage devices However, NVMs often suffer from limited write endurance, a typical PCM cell can only sustain 107-108 writes. Inline deduplication is a promising approach for reducing the storage consumption and the wear of NVM devices. Unfortunately, existing inline deduplication schemes are not efficient for new NVM devices. Since previous data deduplication techniques are designed for block devices such as HDD and SSD, they suffer from high computation and storage overhead due to data chunking and indexing. For example, to chunk 1 TB data without any redundancy using 4 KB blocks, Content-Defined Chunking (CDC) algorithm has to compute rolling hashes for almost 1012 times during the chunking stage. Moreover, the SHA-1 algorithm introduces 5 GB fingerprints (20 bytes per fingerprint) to index those chunks. Since DRAM resource is expensive and limited, many recent proposals store most fingerprints on disk. Thus, it is expensive to search those fingerprints in storage. Moreover, existing inline deduplication schemes have to make a tradeoff between the deduplication ratio and the read throughput. Since the performance difference between the sequential read and the random read is significant for SSD/HDD devices, as shown in Table I, using big chunks (such as 4 KB or 8 KB)

can mitigate the performance degradation due to random accesses to SSD/HDD, and also avoids read amplification if the chunk is larger than the disk block. Moreover, using big chunks can also mitigate the total size of fingerprints and the cost of indexing. In contrast, using small chunks can improve the deduplication ratio, but lowers the read throughput and increase the cost of indexing. Currently, most deduplication systems tend to use big chunks to improve the read throughput and to mitigate the cost of indexing, at the expense of a lower deduplication ratio.

1.1 OBJECTIVE

The objective of the ICID (I/O Causality based In-line Deduplication) project is to pioneer a novel approach to deduplication tailored specifically for non-volatile memory (NVM)-based storage systems, such as Intel Optane DC Persistent Memory Modules. The primary goal is to overcome the limitations of traditional deduplication methods that are inefficient and impractical for NVM due to their reliance on chunking and offline processing. ICID aims to achieve this by introducing an inline deduplication mechanism that operates concurrently with data writes, leveraging the byte-addressability feature of NVM for fine-grained deduplication. Key objectives include developing and implementing a causality-based approach using a B-tree structure to record memory-copy operations in real-time, ensuring that deduplication decisions are based on the order of data accesses. This approach not only maximizes deduplication ratios but also minimizes performance overhead, thereby improving overall system efficiency and responsiveness.

1.2 FEASIBILITY STUDY

The feasibility study shows that this project is technically, economically, operationally, legally, and timely feasible. It provides a secure, efficient, and practical solution for managing medical data with deduplication in NVM storage systems.

1.2.1 TYPES OF FEASIBILITY STUDY

1. Technical Feasibility

This project uses technologies like Java, MySQL, and Non-Volatile Memory (NVM) along with Elliptic Curve Cryptography (ECC) and Blum-Goldwasser Cryptosystem (BGC). These are modern and reliable technologies that can be easily implemented by the project team. The modules for Admin, Patient, Hospital, and Health Sector are successfully designed and tested using available software and hardware.

2. Economic Feasibility

The project helps reduce storage space by avoiding duplicate data using an efficient deduplication technique. This leads to lower memory usage and longer device lifespan. Since the project is developed using open-source tools, there is no additional cost for software licenses.

3. Operational Feasibility

The system is designed to be simple and user-friendly. Patients, hospitals, and health sectors can easily log in, view or upload data, and share encrypted reports. Since all operations are role-based and well-organized, the system fits well into a real-world healthcare environment.

4. Legal Feasibility

Patient data is sensitive and must be kept private. This project ensures privacy and security by encrypting the medical reports before sharing them. Only authorized users can access the data with a decryption key. This approach follows privacy laws and health data protection standards.

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE SURVEY

LITERATURE SURVEY 1

Title: FFCCD: Fence-free crash consistent concurrent defragmentation for persistent memory

Author: Xu, C. Ye, Y. Solihin, and X. Shen.

Year: 2022.

Description:

Persistent Memory (PM) is increasingly supplementing or substituting DRAM as main memory. Prior work have focused on reusability and memory leaks of persistent memory but have not addressed a problem amplified by persistence, persistent memory fragmentation, which refers to the continuous worsening of fragmentation of persistent memory throughout its usage. This paper reveals the challenges and proposes the first systematic crash-consistent solution, Fence-Free Crash-consistent Concurrent Defragmentation (FFCCD). FFCCD resues persistent pointer format, root nodes and typed allocation provided by persistent memory programming model to enable concurrent defragmentation on PM. FFCCD introduces architecture suppor`t for concurrent defragmentation that enables a fence-free design and fast read barrier, reducing two major overheads of defragmenting persistent memory. The techniques is effective (28–73% fragmentation reduction) and fast (4.1% execution time overhead).

LITERATURE SURVEY 2

Title: UCat: Heterogeneous memory management for unikernels.

Author: C. Tian, H. Liu, X. Liao, and H. Jin.

Year: 2022.

Description:

Unkennels provide an efficient and lightweight way to deploy cloud computing services in application-specialized and single-address-space virtual machines (VMs). They can efficiently deploy hundreds of unikernel-based VMs in a single physical server. In such a cloud computing platform, main memory is the primary bottleneck resource for high-density application deployment. Recently, non-volatile memory (NVM) technologies has become increasingly popular in cloud data centres because they can offer extremely large memory capacity at a low expense. However, there still remain many challenges to utilize NVMs for unikernel-based VMs, such as the difficulty of heterogeneous memory allocation and high-performance overhead of address translations.

In this paper, we present UCat, a heterogeneous memory management mechanism that support multi-grained memory allocation for unikernels. We propose front-end/back-end cooperative address space mapping to expose the host memory heterogeneity to unikernels. UCat exploits large pages to reduce the cost of two-layer address translation in virtualization environments, and leverages slab allocation to reduce memory waste due to internal memory fragmentation. We implement UCat based on a popular unikernel—OSv and conduct extensive experiments to evaluate its efficiency. Experimental results show that UCat can reduce the memory consumption of unikernels by 50% and TLB miss rate by 41%, and improve the throughput of real-world benchmarks such as memslap and YCSB by up to 18.5% and 14.8%, respectively.

LITERATURE SURVEY 3

Title: Pattern-guided file compression with user-experience enhancement for log-structured file system on mobile devices.

Author: C. Ji et al.

Year: 2021.

Description:

Mobile applications exhibit unique file access patterns, often involving random accesses of write-mostly files and read-only files. The high write stress of mobile applications significantly impacts on the lifespan of flash-based mobile storage. To reduce write stress and save space without sacrificing user-perceived latency, this study introduces FPC, file access pattern guided compression. FPC is optimized for the random-writes and fragmented-reads of mobile applications. It features dual-mode compression: Foreground compression handles write-mostly files for write stress reduction, while background compression packs random-reading file blocks for boosted read performance. FPC exploits the out-of-place updating design in F2FS, a log-structured file system for mobile devices, for the best effect of the proposed dual-mode compression. Experimental results showed that FPC reduced the volume of total write traffic and executable file size by 26.1% and 23.7% on average, respectively, and improved the application launching time by up to 14.8%.

LITERATURE SURVEY 4

Title: HMvisor: Dynamic hybrid memory management for virtual machines.

Author: D. Yang, H. Liu, H. Jin, and Y. Zhang.

Year: 2021

Description:

Emerging non-volatile memory (NVM) technologies promise high density, low cost and dynamic random access memory (DRAM)-like performance, at the expense of limited write endurance and high write energy consumption. It is more practical to use NVM combining with the traditional DRAM. However, the hybrid memory management such as page migration becomes more challenging in a virtualization environment because virtual machines (VMs) are unaware of the memory heterogeneity. In this paper, we propose HMvisor, a hypervisor and VM coordinated hybrid memory management mechanism to better utilize DRAM and NVM resources. HMvisor exposes the memory heterogeneity to VMs by mapping virtual NUMA nodes to different physical NUMA nodes. We propose a lightweight and efficient page migration mechanism by decoupling page hotness tracking from page migration. HMvisor performs those operations in the hypervisor and VMs separately, without disrupting the execution of VMs. We also propose a memory resource trading policy to adjust the capacity of DRAM and NVM for each VM, with the monetary cost unchanged. We implement our prototype system based on QEMU/KVM and evaluate it with several benchmarks. Experimental results show that HMvisor can reduce 50% of write traffic to NVM with less than 5% performance overhead. Moreover, the hybrid memory adjustment scheme in HMvisor can significantly improve application performance by up to 30 ×

LITERATURE SURVEY 5

Title: "NV-Dedup: High-performance inline deduplication for non-volatile memory.

Author: C. Wang, Q. Wei, J. Yang, C. Chen, Y. Yang, and M. Xue,

Year: 2018.

Description:

The byte-addressable non-volatile memory (NVM) is a promising medium for data storage. NVM-oriented file systems have been designed to explore NVM's performance potential. Meanwhile, applications may write considerable duplicate data. For NVM, a removal of duplicate data can promote space efficiency, improve write endurance, and potentially improve the performance by avoidance of repeatedly writing the same data. However, we have observed severe performance degradations when implementing a state-of-the-art inline deduplication algorithm in an NVM-oriented file system. A quantitative analysis reveals that, with NVM, 1) the conventional way to manage deduplication metadata for block devices, particularly in light of consistency, is inefficient, and, 2) the performance with deduplication becomes more subject to fingerprint calculations. We hence propose a deduplication algorithm called NV-Dedup. NV-Dedup manages deduplication metadata in a fine-grained, CPU and NVM-favored way, and preserves the metadata consistency with a lightweight transactional scheme. It also does workload-adaptive fingerprinting based on an analytical model and a transition scheme among fingerprinting methods to reduce calculation penalties. We have built a prototype of NV-Dedup in the Persistent Memory File System (PMFS). Experiments show that, NV-Dedup not only substantially saves NVM space, but also boosts the performance of PMFS by up to 2.1x.

2.2 SCOPE OF PROJECT

In order to improve the effectiveness and performance of deduplication technology in non-volatile memory (NVM)-based storage systems, the ICID (I/O Causality based In-line Deduplication) project's scope includes multiple important aspects. In order to achieve finer granularity in data deduplication, it first seeks to improve and optimize deduplication algorithms designed specifically for NVM by utilizing its byte-addressability capability. Creating data structures and algorithms, like the causality-based B-tree technique, is one way to effectively handle memory-copy operations inline during data writes. Second, the project enhances data localization and memory management techniques inside the deduplication framework to solve the scalability issues present in large-scale storage environments.

2.2.1 PROBLEM STATEMENT

The problem addressed by this project Managing memory-copy operations is crucial for optimizing data handling and performance in various computing processes. This involves strategically organizing data to minimize unnecessary copying while ensuring efficient access and modification. Techniques such as using pointers instead of duplicating data, employing memory pools, and implementing efficient algorithms can significantly reduce overhead. Additionally, leveraging batch processing can enhance throughput by reducing the number of individual copy operations. Effective management of memory-copy operations not only improves system performance but also helps in maintaining data integrity and responsiveness, making it essential for applications that rely on high-speed data processing and deduplication tasks.

2.2.2 EXISTING SYSTEM

- Traditional chunk-based offline deduplication methods are not efficient for NVM-based storage due to their inability to fully utilize the byte-addressability feature of NVMs. These methods typically incur significant performance overhead due to chunking and indexing.
- Deduplication processes can be resource-intensive, potentially impacting performance during execution.
- Regular deduplication requires ongoing maintenance and monitoring, which can increase administrative workload.

2.2.3 DISADVANTAGES OF EXISTING SYSTEM

- Increased Memory Consumption.
- High overhead from memory-copy operations can slow down processing, especially in high-volume environments.
- Continuous memory allocation and deallocation may lead to fragmentation, reducing available memory and performance.

2.3 PROPOSED SYSTEM

- DATA deduplication techniques have long been studied for space saving in storage systems. They eliminate duplicate data at file or chunk levels by identifying the same data using cryptographic hash functions such as SHA-1 or MD5.
- In this paper, we propose I/O Causality based In-line Deduplication (ICID) to maximize the deduplication ratio for NVM-based storage systems. Unlike previous inline deduplication schemes that use hash indexes to identify duplicate data slices, ICID records memorycopy operations in a B-tree structure to achieve causality-based inline deduplication.
- In contrast, the inline deduplication detects the duplicate data before writing it back into storage, and thus can reduce data redundancy and the wear of storage devices.

2.3.1 PROPOSED SYSTEM ADVANTAGES

- Maximized Deduplication Ratio.
- Improved Performance.
- Efficient Memory Management.
- Causality-based Approach.

2.4 REQUIREMENTS

We can see from the results that on each database, the error rates are very low due to the discriminatory power of features and the regression capabilities of classifiers. Comparing the highest accuracies (corresponding to the lowest error rates) to those of previous works, our results are very competitive.

2.4.1 SOFTWARE REQUIREMENTS

The software requirements document is the specification of the system. It should include both a definition and a specification of requirements. It is a set of what the system should do rather than how it should do it. The software requirements provide a basis for creating the software requirements specification. It is useful in estimating cost, planning team activities, performing tasks and tracking the teams and tracking the team's progress throughout the development activity.

- Front End : J2EE (JSP, SERVLET)
- Back End : MY SQL 5.5
- Operating System : Windows 11
- IDE : Eclipse

2.4.2 HARDWARE REQUIREMENTS

The hardware requirements may serve as the basis for a contract for the implementation of the system and should therefore be a complete and consistent specification of the whole system. They are used by software engineers as the starting point for the system design. It shoulds what the system do and not how it should be implemented.

- PROCESSOR : AMD RYZEN 5.
- RAM : 8GB DD RAM
- MONITOR : 15.6” COLOR
- HARD DISK : 500 GB

2.4.3 FUNCTIONAL REQUIREMENTS

A functional requirement defines a function of a software-system or its component. A function is described as a set of inputs, the behavior, Firstly, the system is the first that achieves the standard notion of semantic security for data confidentiality in attribute-based deduplication systems by resorting to the hybrid cloud architecture.

2.4.4 NON-FUNCTIONAL REQUIREMENTS

The major non-functional Requirements of the system are as follows

- **Usability**

The system is designed with completely automated process hence there is no or less user intervention.

- **Reliability**

The system is more reliable because of the qualities that are inherited from the chosen platform java. The code built by using java is more reliable.

- **Performance**

This system is developing in the high level languages and using the advanced front-end and back-end technologies it will give response to the end user on client system with in very less time.

- **Supportability**

The system is designed to be the cross platform supportable. The system is supported on a wide range of hardware and any software platform, which is having JVM, built into the system.

- **Implementation**

The system is implemented in web environment using struts framework. The apache tomcat is used as the web server and windows xp professional is used as the platform. Interface the user interface is based on Struts provides HTML Tag

2.5 METHODOLOGY

The ICID (I/O Causality based In-line Deduplication) project introduces a cutting-edge approach to data deduplication designed specifically for non-volatile memory (NVM)-based storage systems. Unlike traditional methods that rely on chunking and offline processing, ICID implements deduplication inline during data writes, leveraging the byte-addressability feature of NVM for precise and efficient deduplication. Central to ICID is its causality-based approach using a B-tree structure, which records memory-copy operations in real-time.

2.5.1 MODULES NAMES:

- User Interface Design
- Admin
- Hospital
- Health Sector
- Patient/User
- Encryption

2.5.2 MODULES EXPLANATION

2.5.2.1 User Interface Design

To connect with server user must give their username and password then only they can able to connect the server. If the user already exits directly can login into the server else user must register their details such as username, password, Email id, City and Country into the server. Database will create the account for the entire user to maintain upload and download rate. Name will be set as user id. Logging in is usually used to enter a specific page. It will search the query and display the query.

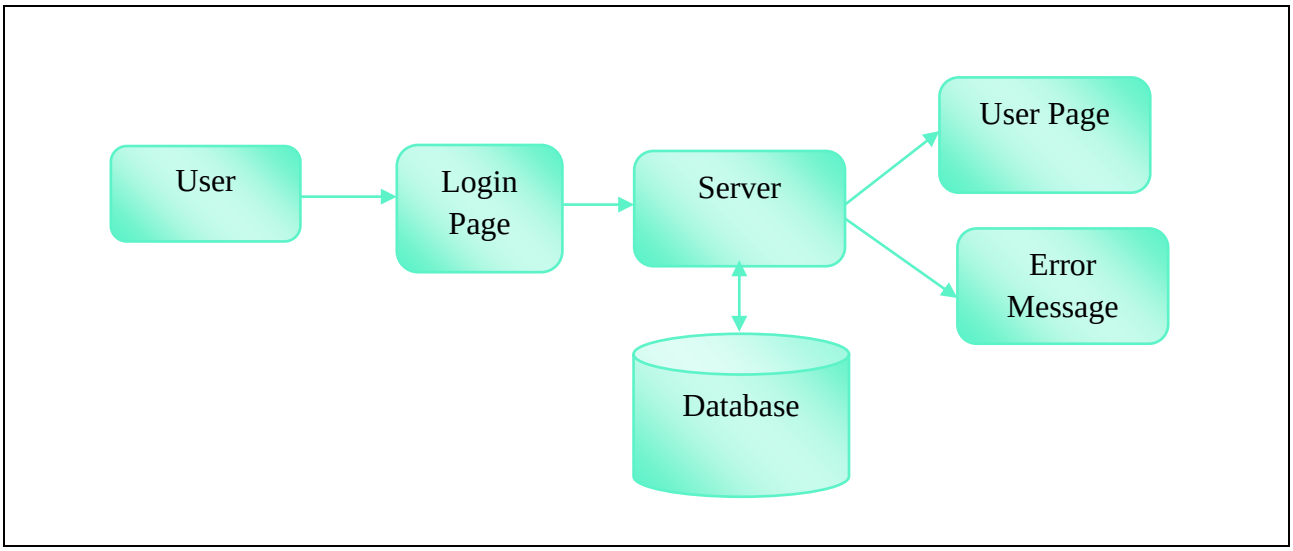


FIGURE 2.5.2.1: User Interface Design

2.5.2.2 Hospital

Hospital is the second module in our project, Hospital first login with Username and password then view patient details and symptoms or request and approve the patient and join. After hospital management accepting the patient, he will view send the symptoms to health sector for scanning the patient and that data will be send to Hospital. Hospital management will send the reports to patient in encrypted format for security purpose. Patient have to request hos management and send keys to decrypt the reports.

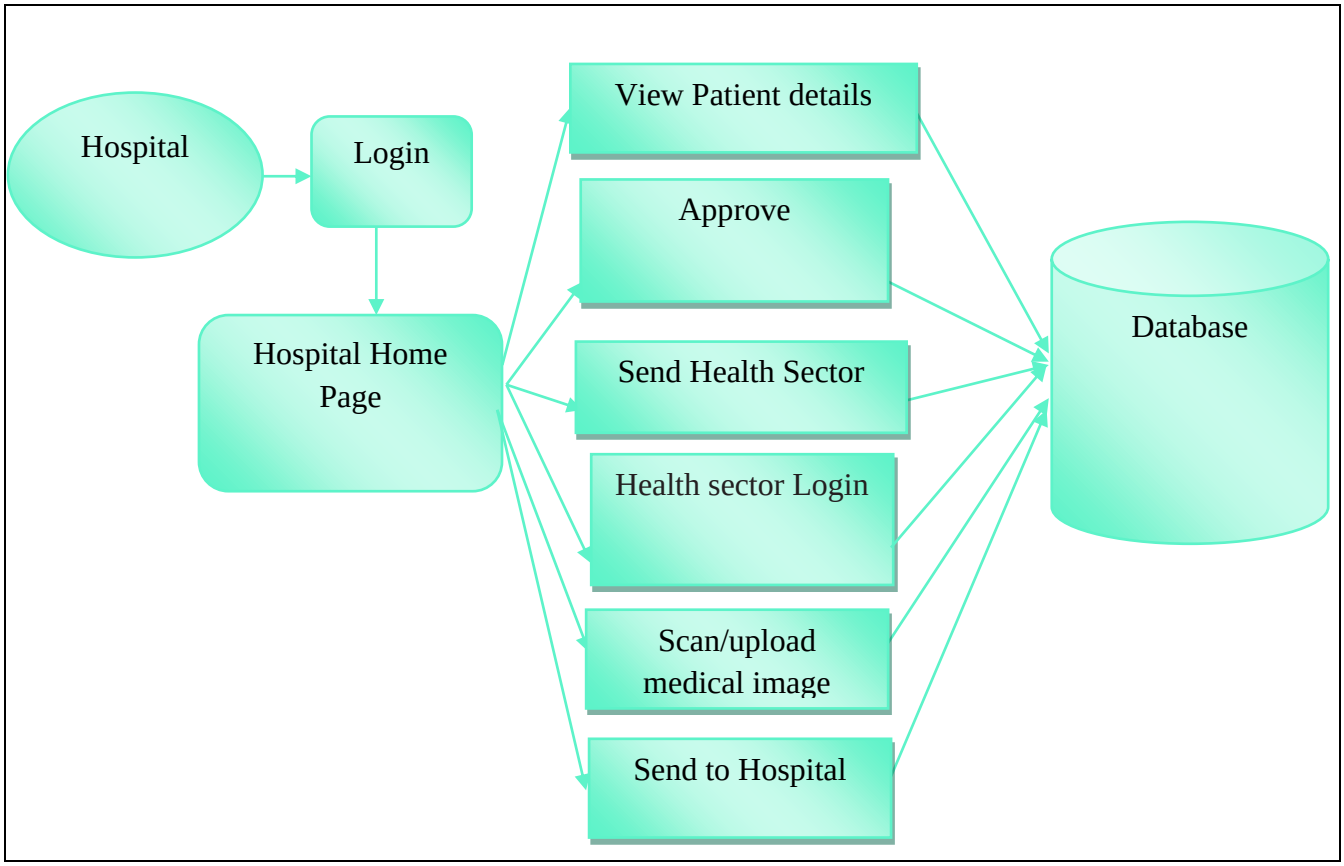


FIGURE 2.5.2.2: Hospital Module

2.5.2.3 Admin

This is the third module in our project where Admin Controls all the of the project role. Administration managers ensure an organization's activities run smoothly and efficiently. The primary goals of an administration manager are to direct, control and supervise the support services of the organization to facilitate its success.

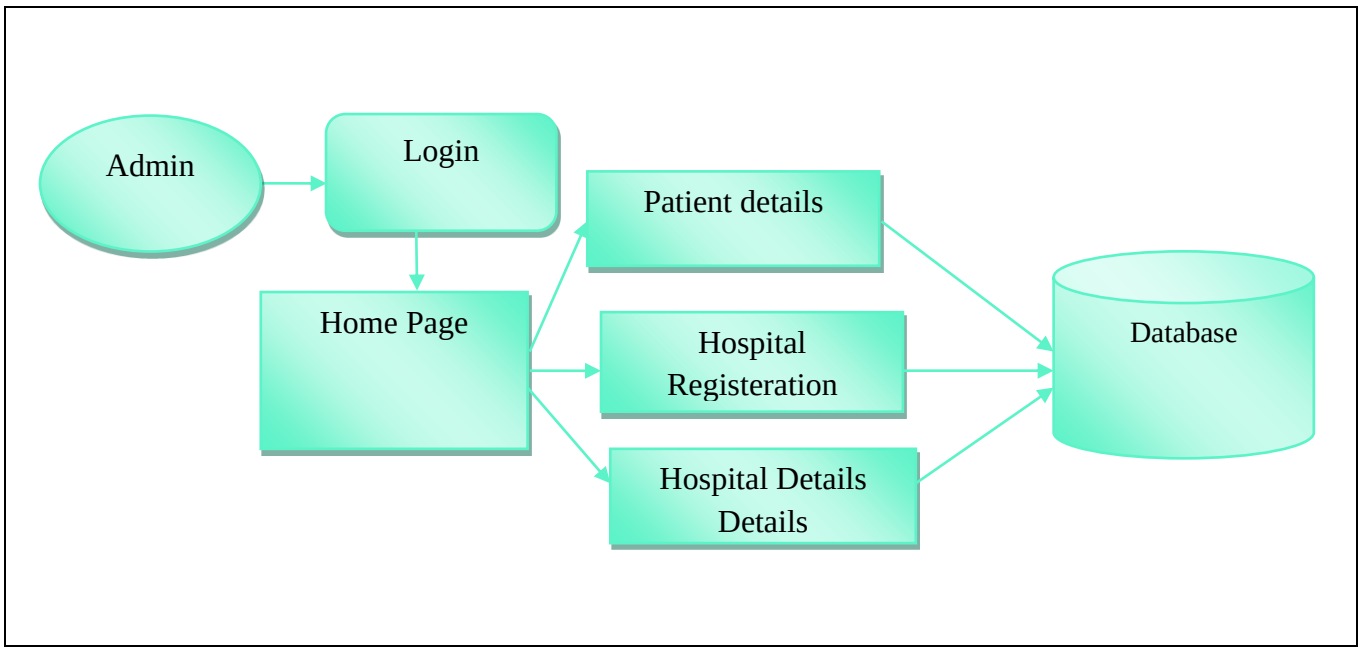


FIGURE 2.5.2.3: Admin Module

2.5.2.4 Health Sector

This is the fourth module in our project is Health Sector. Here Health Sector view patient symptoms where hospital management has sent. Based on patient data he will scan the patient and those reports as sent to Hospital Management.

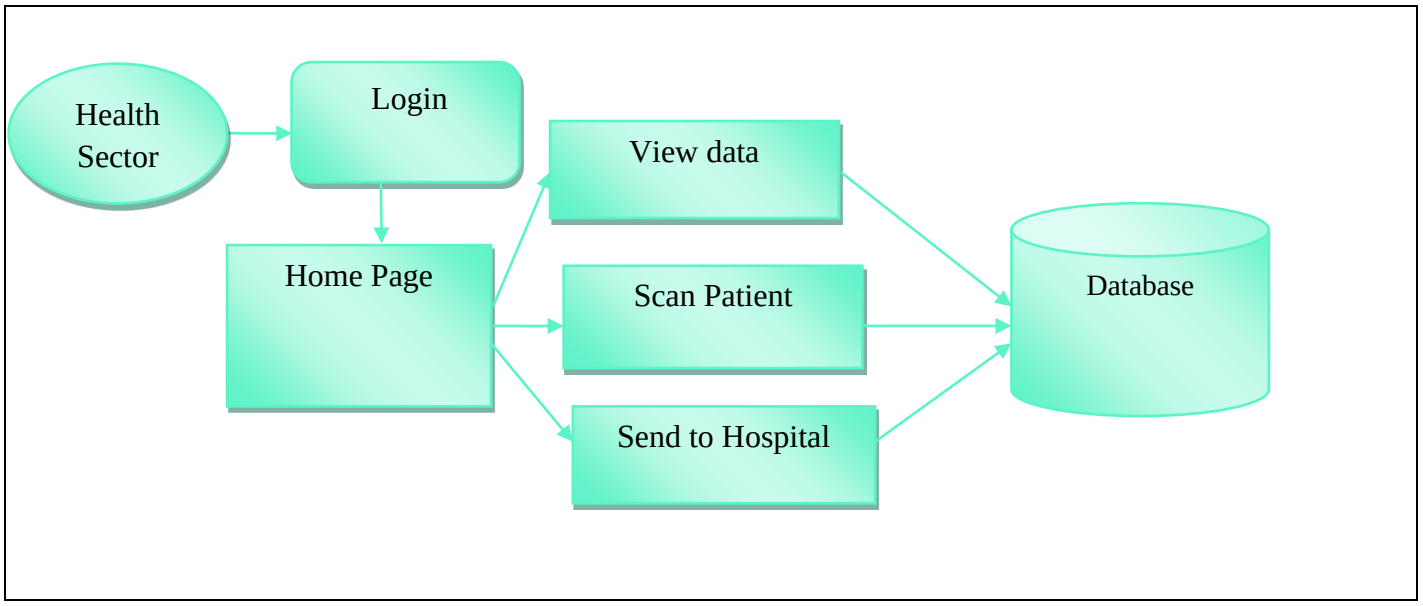


FIGURE 2.5.2.4: Health Sector Module

2.5.2.5 Encryption

The encryption of medical images using Elliptic Curve Cryptography (ECC) and the Blum-Goldwasser cryptosystem provides a robust framework for ensuring the confidentiality and integrity of sensitive medical data. ECC offers a high level of security with smaller key sizes, making it ideal for resource-constrained environments typical in medical applications.

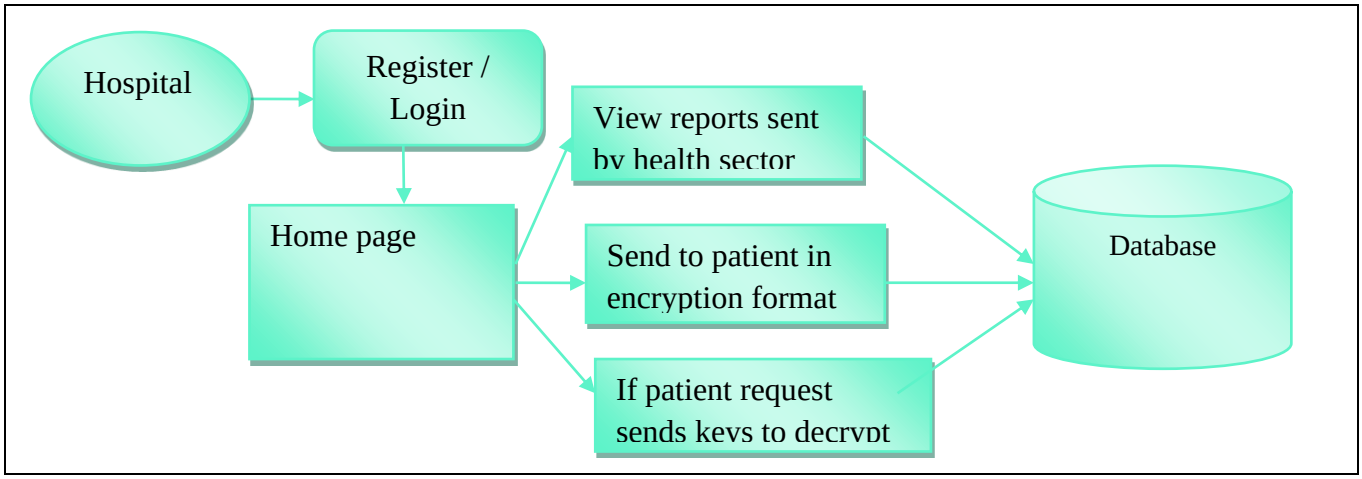


FIGURE 2.5.2.5: Encryption Module

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2.6. TECHNIQUES OR ALGORITHMS

2.6.1 Elliptic Curve Cryptography (ECC):

Elliptic Curve Cryptography (ECC) is a powerful encryption technique that utilizes the mathematical properties of elliptic curves over finite fields to provide secure communication. ECC is particularly valued for its ability to deliver strong security with smaller key sizes compared to traditional cryptographic methods like RSA, making it efficient and suitable for resource-constrained environments. This efficiency leads to faster computations and reduced storage requirements, which are essential for applications in mobile devices and embedded systems. ECC is widely used in various security protocols, including SSL/TLS for secure web communications, digital signatures, and key exchange mechanisms, ensuring robust protection for sensitive data and communications in an increasingly interconnected digital landscape.

2.6.2 Blum-Goldwasser Cryptosystem (BGC):

The Blum-Goldwasser Cryptosystem (BGC) is a probabilistic public-key encryption scheme that is founded on the difficulty of factoring large composite numbers. Developed by Manuel Blum and Shafi Goldwasser, BGC is notable for its strong theoretical security properties, as it produces ciphertext that is indistinguishable from random noise. The system operates using a pair of large prime numbers to generate a modulus, and it relies on the quadratic residues of these primes to encrypt messages. This unique approach allows BGC to provide a robust level of security while maintaining simplicity in its design. Due to its security foundations, the Blum-Goldwasser Cryptosystem is particularly suitable for applications requiring high assurance in confidentiality and integrity, making it an essential tool in the field of cryptography.

2.6.3 Hash-Based Data Deduplication with Offline Processing

The existing system uses a traditional chunk-based deduplication method, where data is broken into fixed-size chunks (e.g., 4 KB) and each chunk is hashed using algorithms like SHA-1 or MD5. These hashes are stored as fingerprints in an index, and duplicates are detected by comparing new chunk hashes with existing ones.

However, this deduplication is performed offline, meaning after the data has already been written to storage. This causes write amplification, increases storage overhead, and delays the deduplication process. The large fingerprint indexes also consume significant memory and reduce system speed, especially in large-scale environments.

CHAPTER 3

DESIGN AND DEVELOPMENT

3.1 SYSTEM ARCHITECTURE

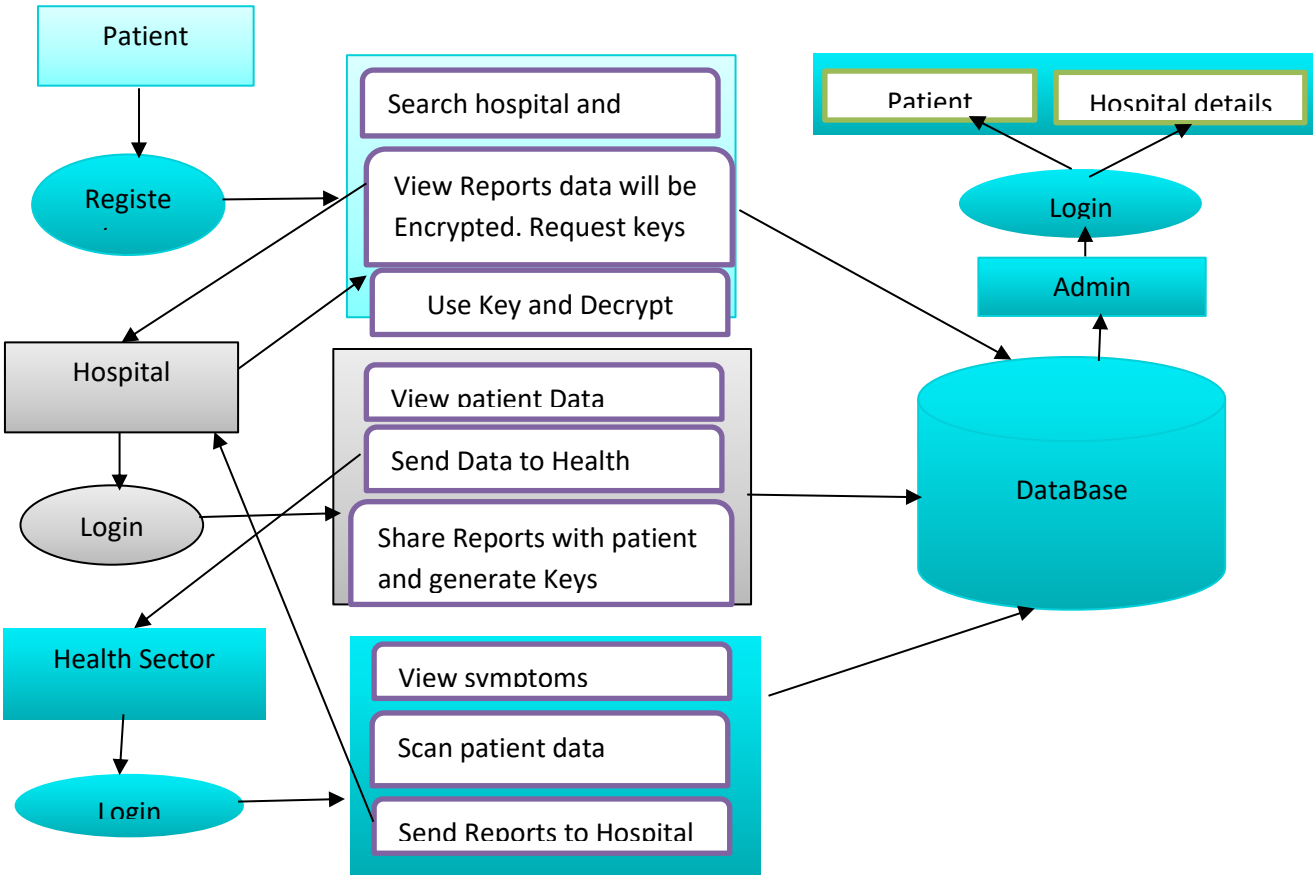


FIGURE 3.1: System Architecture

EXPLANATION

This diagram explains how a digital healthcare system connects patients, hospitals, and the health sector to manage and share medical data securely. Patients start by registering or logging in, then they can search for hospitals and view their health reports. These reports are encrypted for security, and patients need a key to decrypt and read them. This ensures that only the patient can access their private medical information.

3.2 UML DIAGRAMS

GENERAL

Design Engineering deals with the various UML [Unified Modelling language] diagrams for the implementation of project. Design is a meaningful engineering representation of a thing that is to be built. Software design is a process through which the requirements are translated into representation of the software. Design is the place where quality is rendered in software engineering. Design is the means to accurately translate customer requirements into finished product.

3.2.1 DATA FLOW DIAGRAM (Level 0):

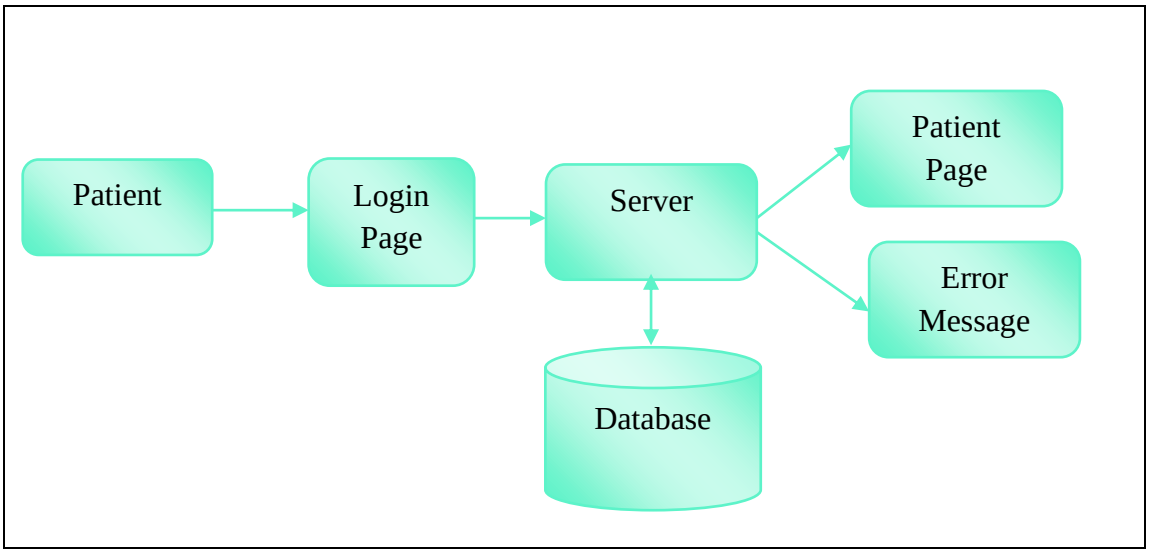


FIGURE 3.2.1:LEVEL 0 DATA FLOW DIAGRAM

Level 1:

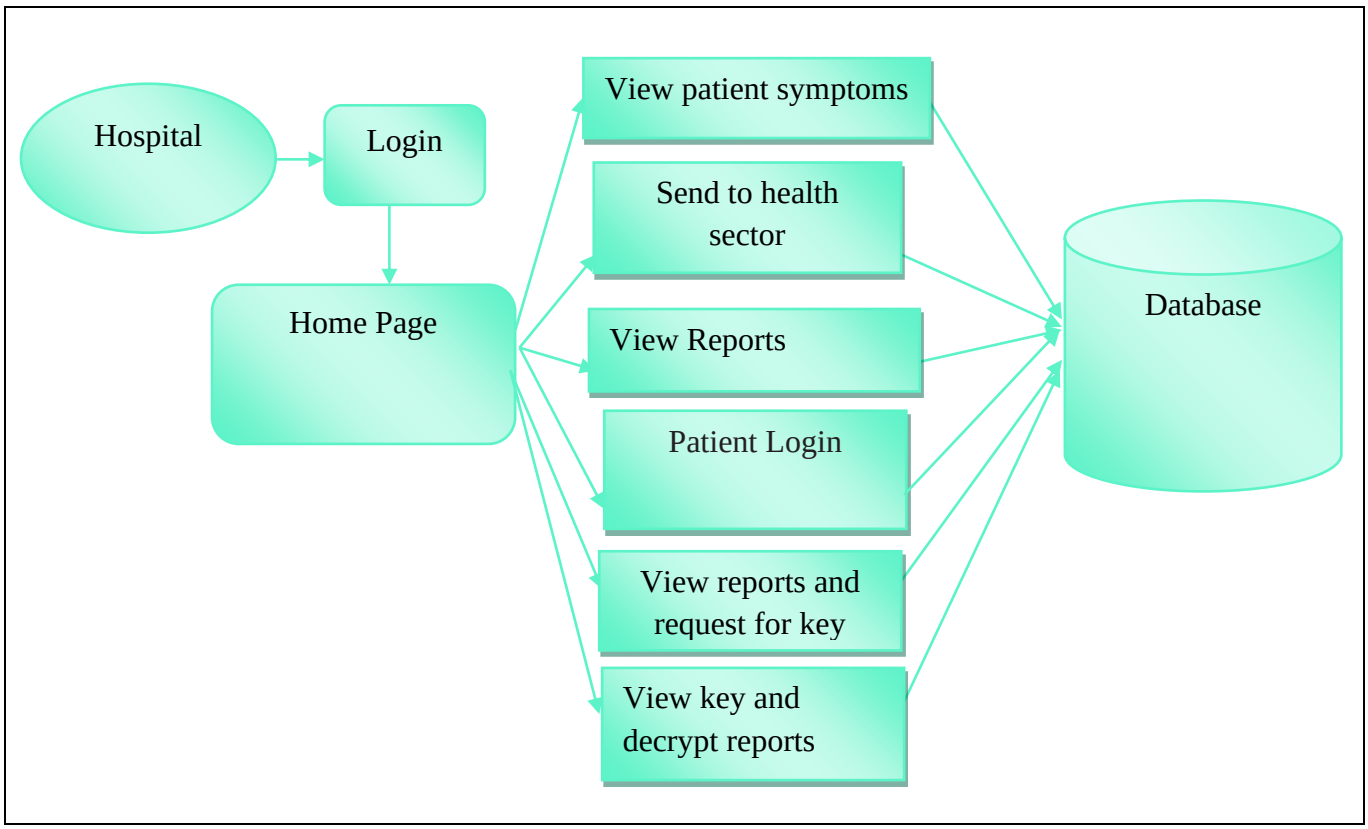


FIGURE 3.2.1: LEVEL 1 DATA FLOW DIAGRAM

EXPLANATION

The Level 1 Data Flow Diagram maps core processes and data exchanges: patients log in, view reports, and request decryption keys; hospitals receive and forward patient data to the health sector; health sector scans and returns analyzed reports; EncryptionService secures deduplicated reports into the database; upon authorization, patients retrieve decryption keys to access their medical data securely, ensuring confidentiality, efficiently.

3.2.2 USE CASE DIAGRAM

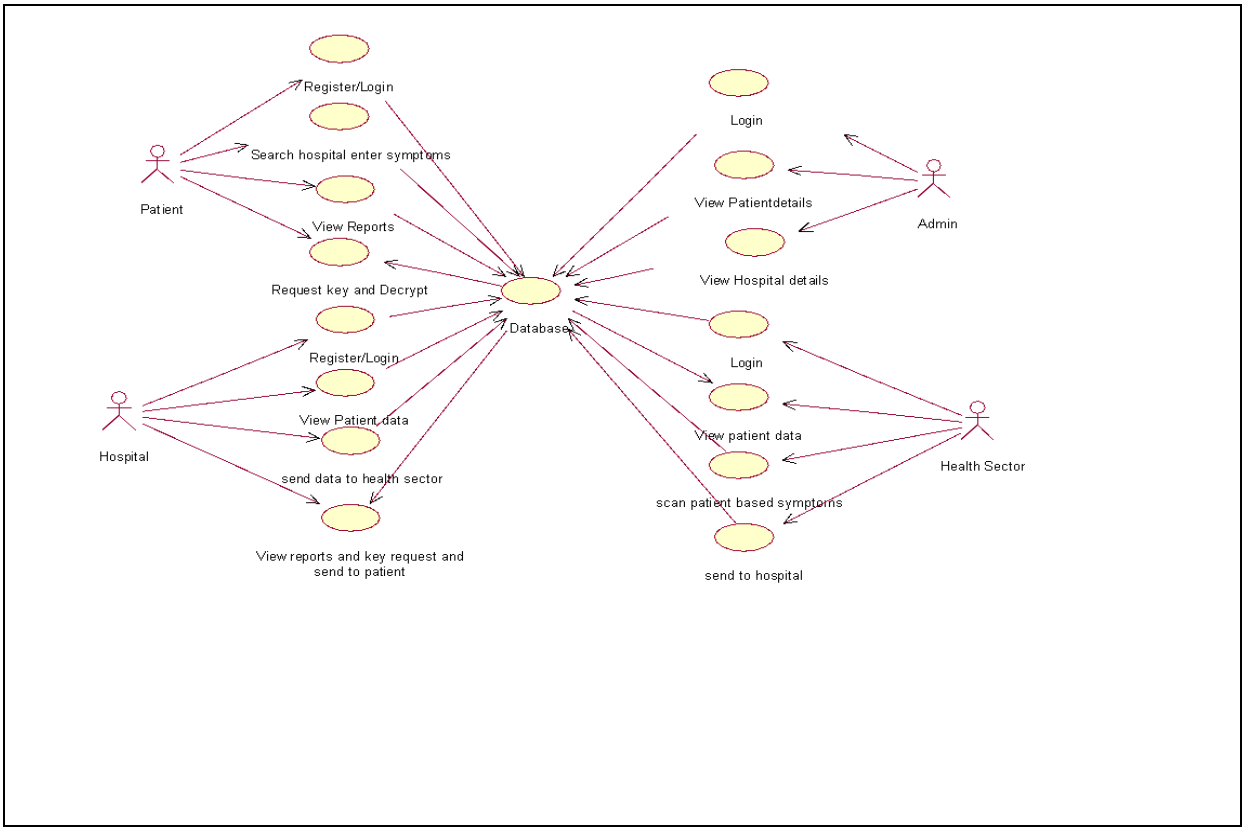


FIGURE 4.1.1 : Use Case Diagram

EXPLANATION:

In this project, the use case diagram captures the functional flow between patients, hospitals, the health sector, and admin. It shows actions like login, data submission, encryption, and report sharing. Each actor interacts with the system based on their role. This helps visualize the main services provided to each stakeholder.

3.2.3 Class Diagram

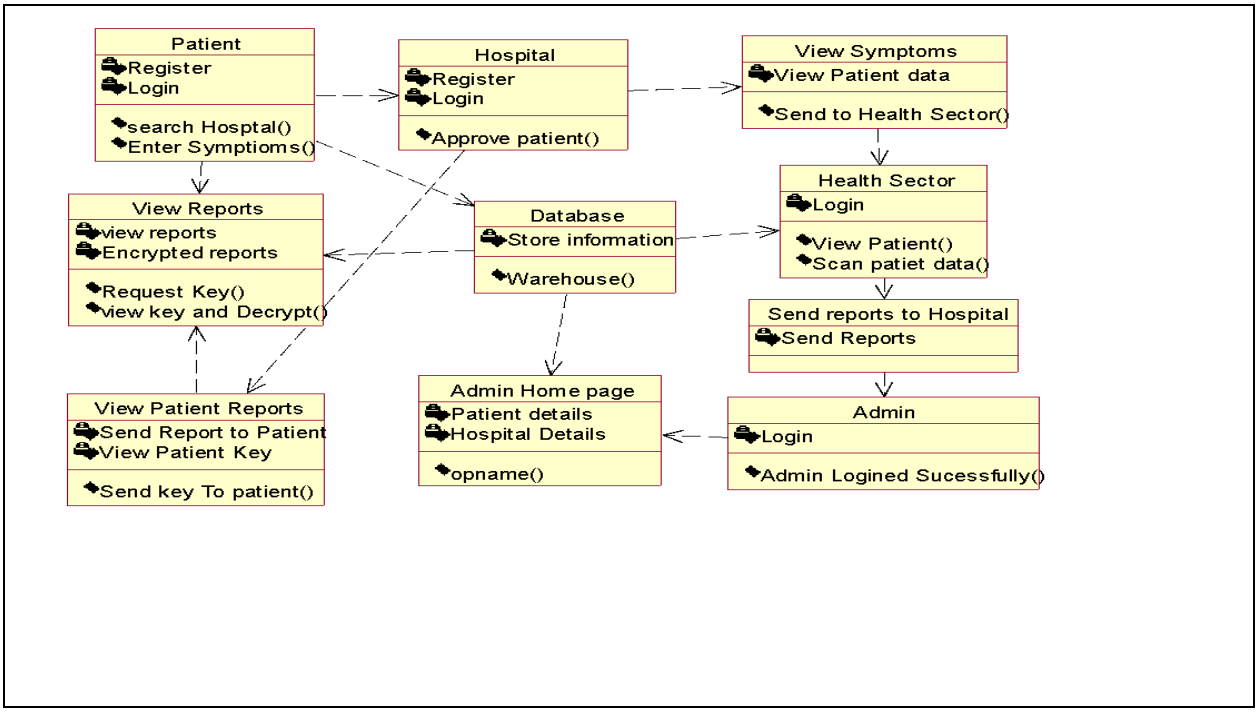


FIGURE 3.2.3 : Class Diagram

EXPLANATION

The class diagram includes Patient, Hospital, HealthSector, Admin, EncryptionService, and Database classes with attributes (userID, credentials, reportData) and methods (login, register, scanReport, encrypt, decrypt, store). Patients authenticate via Admin and submit data to Hospital, which sends it to HealthSector for scanning/deduplication. EncryptionService secures deduplicated reports for Database storage, decrypts them on authorized requests, ensuring efficient, secure NVM-based medical data management.

3.2.4 ACTIVITY DIAGRAM

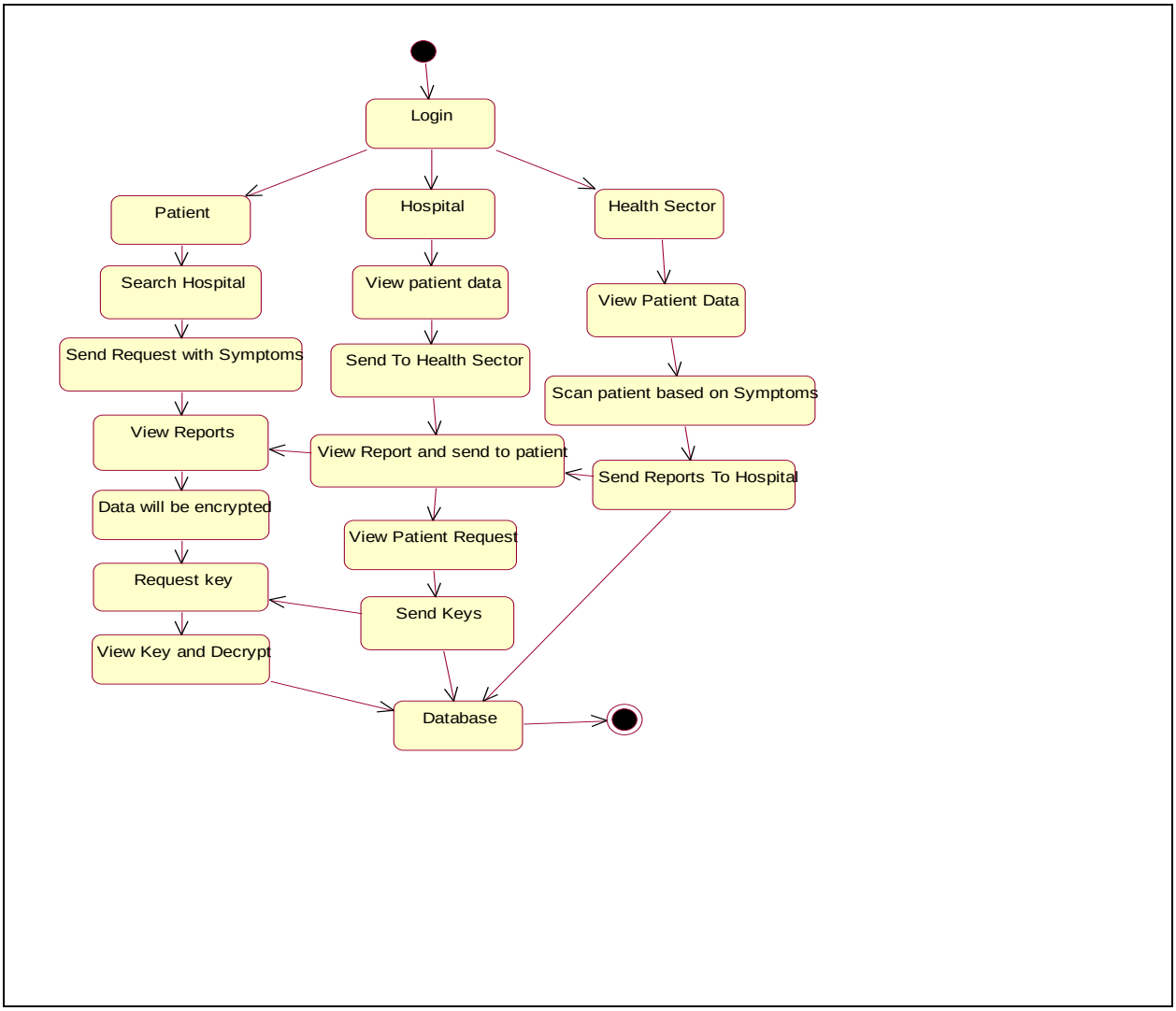


FIGURE 3.2.4: Activity Diagram

EXPLANATION:

This diagram presents the workflow of tasks like patient registration, hospital approval, health sector scanning, and encrypted report distribution. Each activity transitions logically, with conditions and loops depicted clearly. It helps streamline process flows and reduce system bottlenecks in real-world usage.

3.2.5 SEQUENCE DIAGRAM

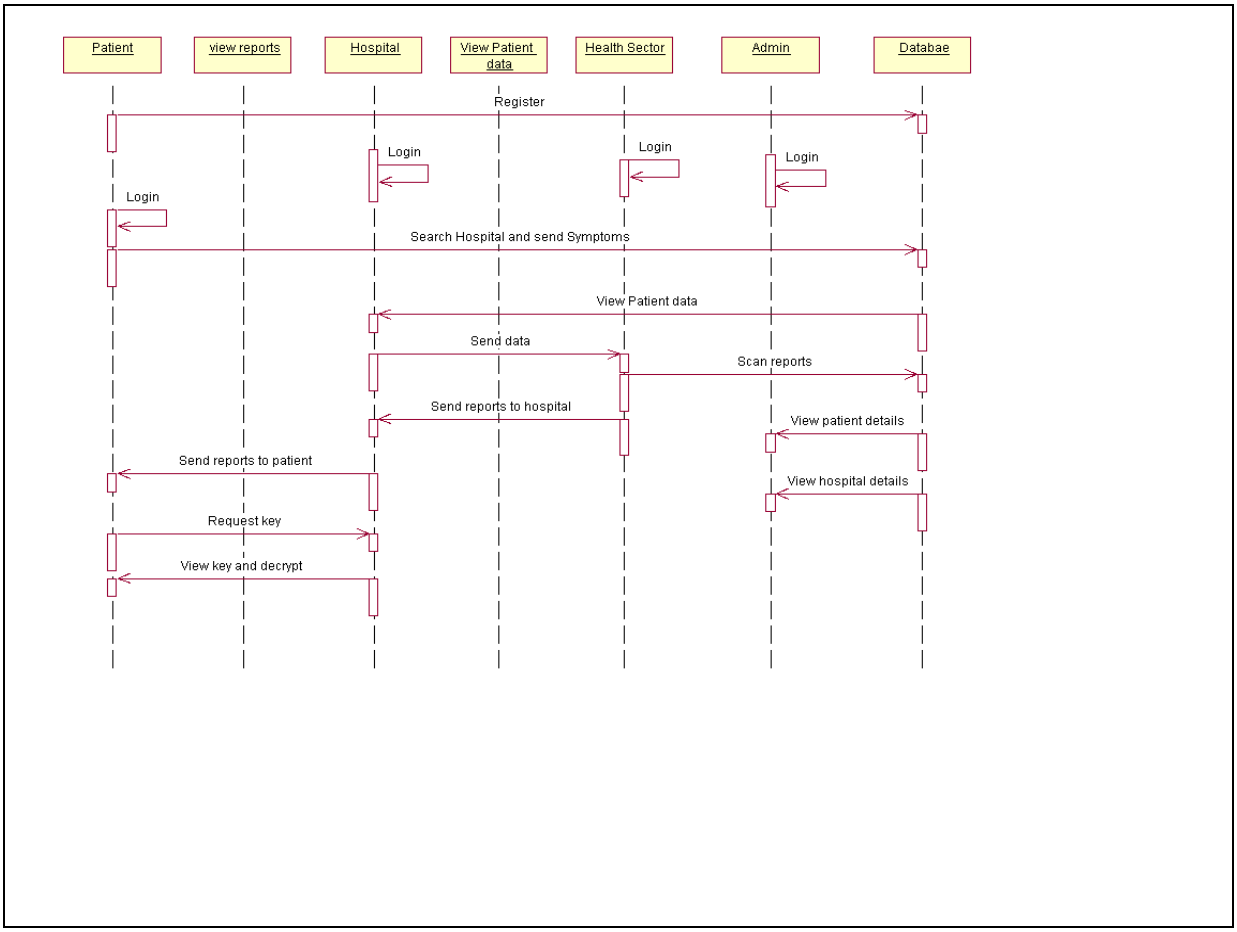


FIGURE 3.2.5: Sequence Diagram

EXPLANATION:

This diagram shows the timeline-based interaction between objects such as Patient, Hospital, Health Sector, and Database. It visualizes how a patient's request travels through the system, how the hospital forwards data, and how encrypted reports are returned. It is crucial to understanding the order and logic of operations in the project.

3.2.6 Collaboration Diagram

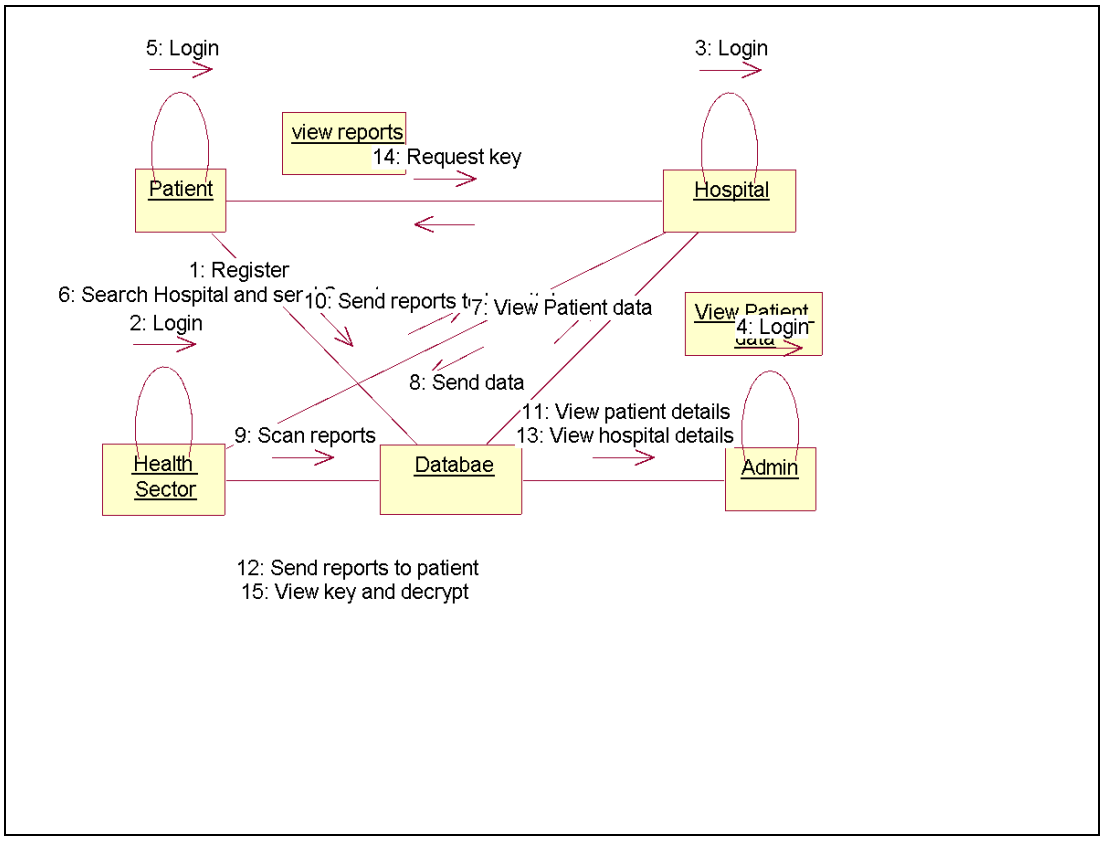


FIGURE 3.2.6 : Collaboration Diagram

EXPLANATION:

The collaboration diagram represents the flow of messages between components while highlighting structural relationships. In this project, it shows how data moves between user modules, encryption logic, and the database. This helps in identifying how tightly or loosely coupled each module is.

3.2.7 Component Diagram

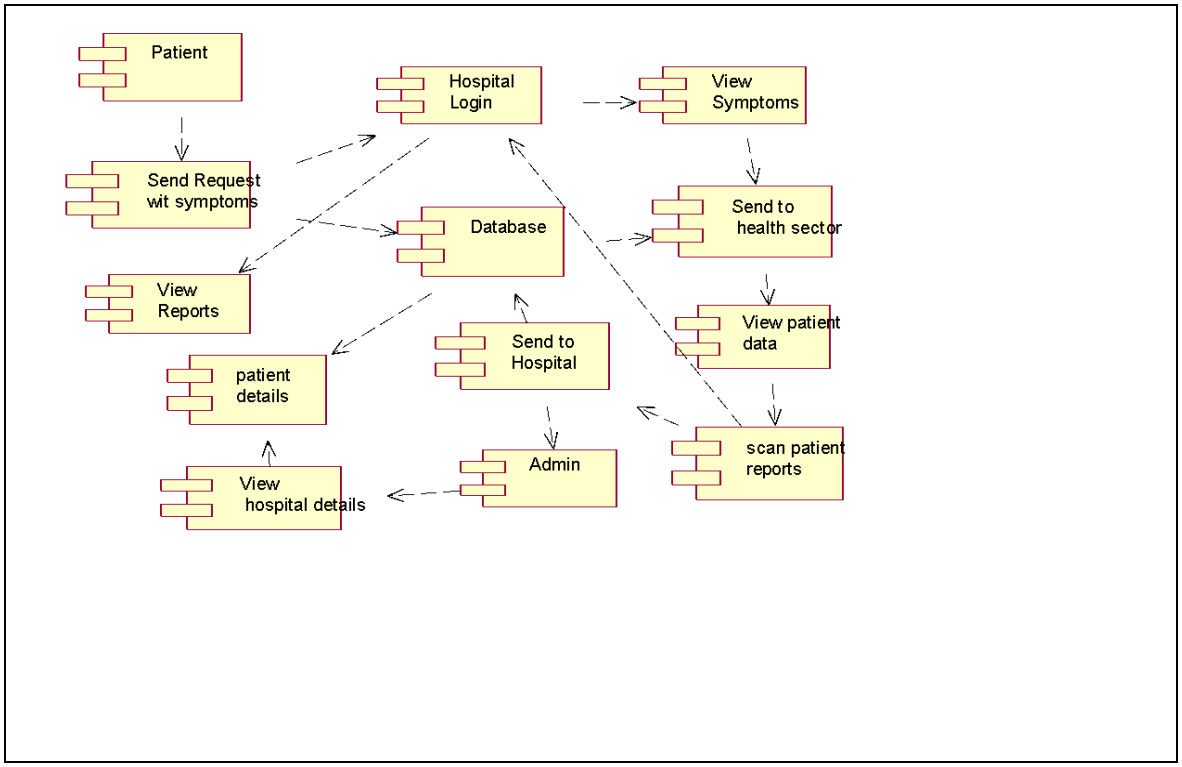


FIGURE 3.2.7 : Component Diagram

EXPLANATION:

Here, the component diagram illustrates how core components—like the login system, encryption engine, and data viewer—are connected. Each component handles a distinct functionality in the web application. The diagram ensures the modules can be independently developed and later integrated efficiently.

3.2.8 Deployment Diagram:

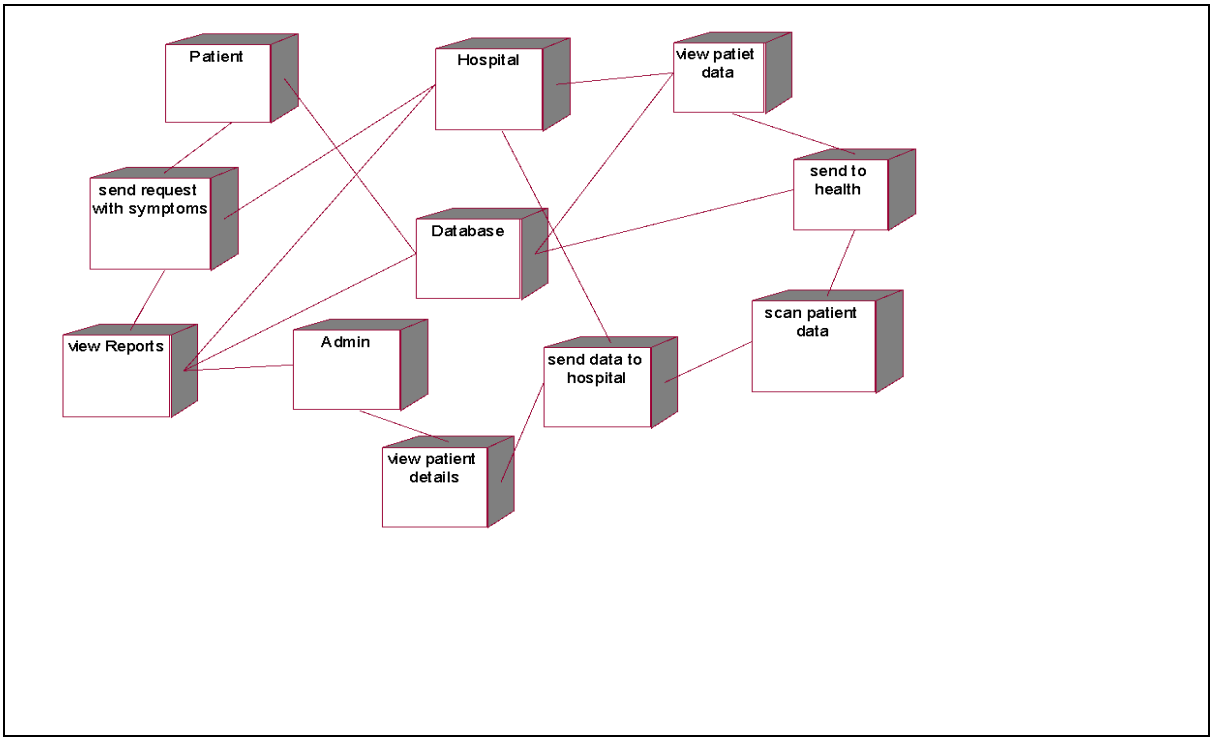


FIGURE 3.2.8 : Deployment Diagram

EXPLANATION:

This diagram maps how software components are deployed across servers, databases, and client systems. In this project, it shows how patients interact through a web interface, while data is processed on servers and stored in the database. It supports real-time deployment decisions and infrastructure planning.

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3.3 SAMPLE CODE

SimilarityValueController.java

```
package controller;
import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import service.AdminManageService;
import service.AdminManageServiceImpl;
@WebServlet("/similarityValue")
public class SimilarityValueController extends HttpServlet {
	@Override
	protected void doGet(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOException {
		doPost(req, resp);
	}
	@Override
	protected void doPost(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOException {
		AdminManageService service = new AdminManageServiceImpl();
		int[] i = service.getSimilarityValue();
		System.out.println(i[0]);//print the value
		if (i != null) {
			int Romance = i[0] * 10;
			int Comedy = i[1] * 10;
```

```
                int Drama = i[2] * 10;
                int Animation = i[3] * 10;
                int Fantasy = i[4] * 10;
                int Action = i[5] * 10;
                req.setAttribute("Romance", Romance);
                req.setAttribute("Comedy", Comedy);
                req.setAttribute("Drama", Drama);
                req.setAttribute("Animation", Animation);
                req.setAttribute("Fantasy", Fantasy);
                req.setAttribute("Action", Action);
            }           req.getRequestDispatcher("similarValue.jsp").forward(req, resp);
        }

}
```

CHAPTER 4

RESULTS & DISCUSSION

4.1 GENERAL

This project is implements like web application using COREJAVA and the Server process is maintained using the SOCKET & SERVERSOCKET and the Design part is played by Cascading Style Sheet.

4.2 VARIOUS SNAPSHOTS

4.2.1 welcome Page

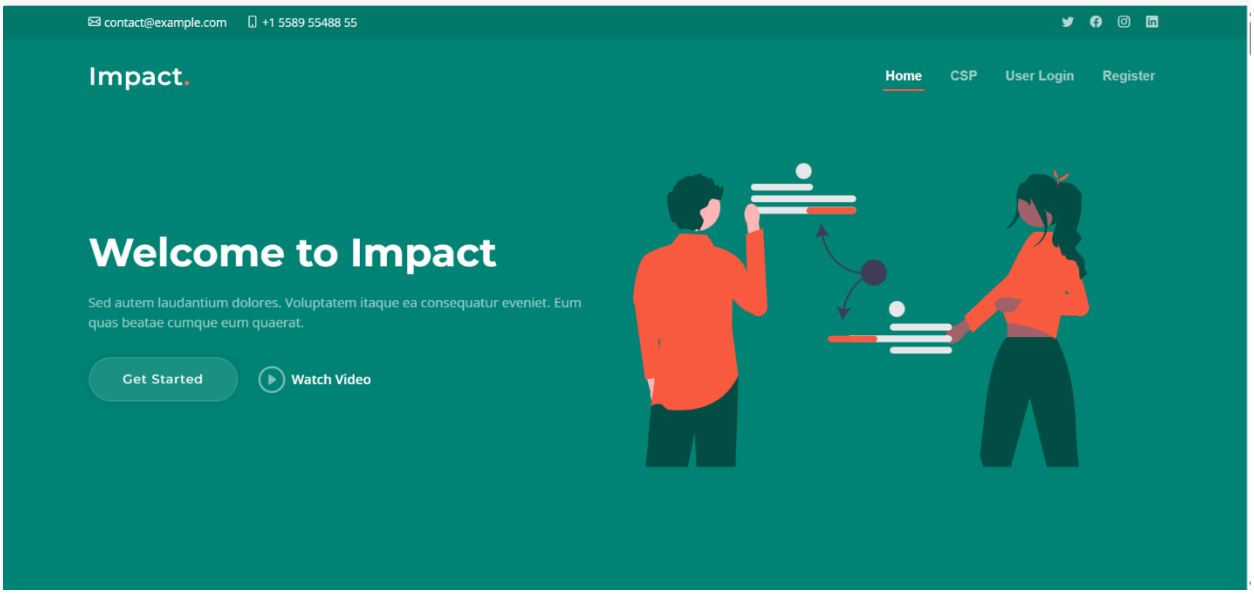


FIGURE 4.2.1: Welcome Page.

This is the Welcome page where users are welcomed and can navigate to core features like registration, login, and starting the application.

4.2.2 User Login Page

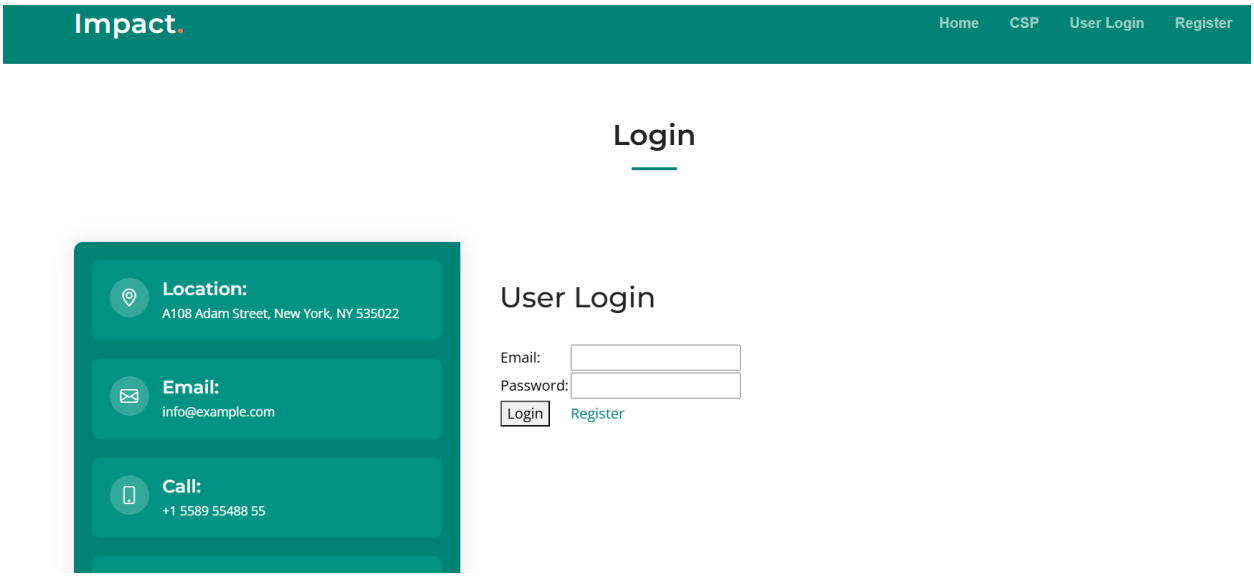


FIGURE 4.2.2: User Login Page

This is the **User Login page** where users enter their email and password to access the system. It also provides contact information and a link to register for new users.

4.2.3 User Registration Page

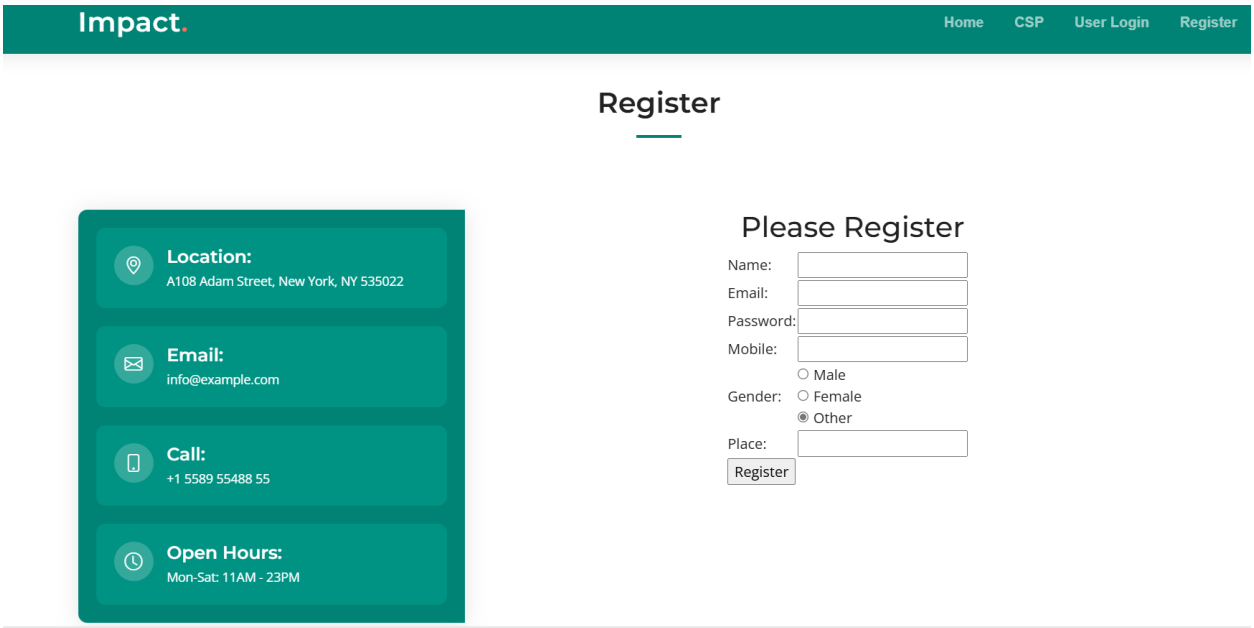


FIGURE 4.2.3: User Registration Page

This page allows new users to create an account by entering personal details such as name, email, mobile, gender, and location.

It ensures authenticated access to the system by capturing user credentials securely.

4.2.4 Data Base Page

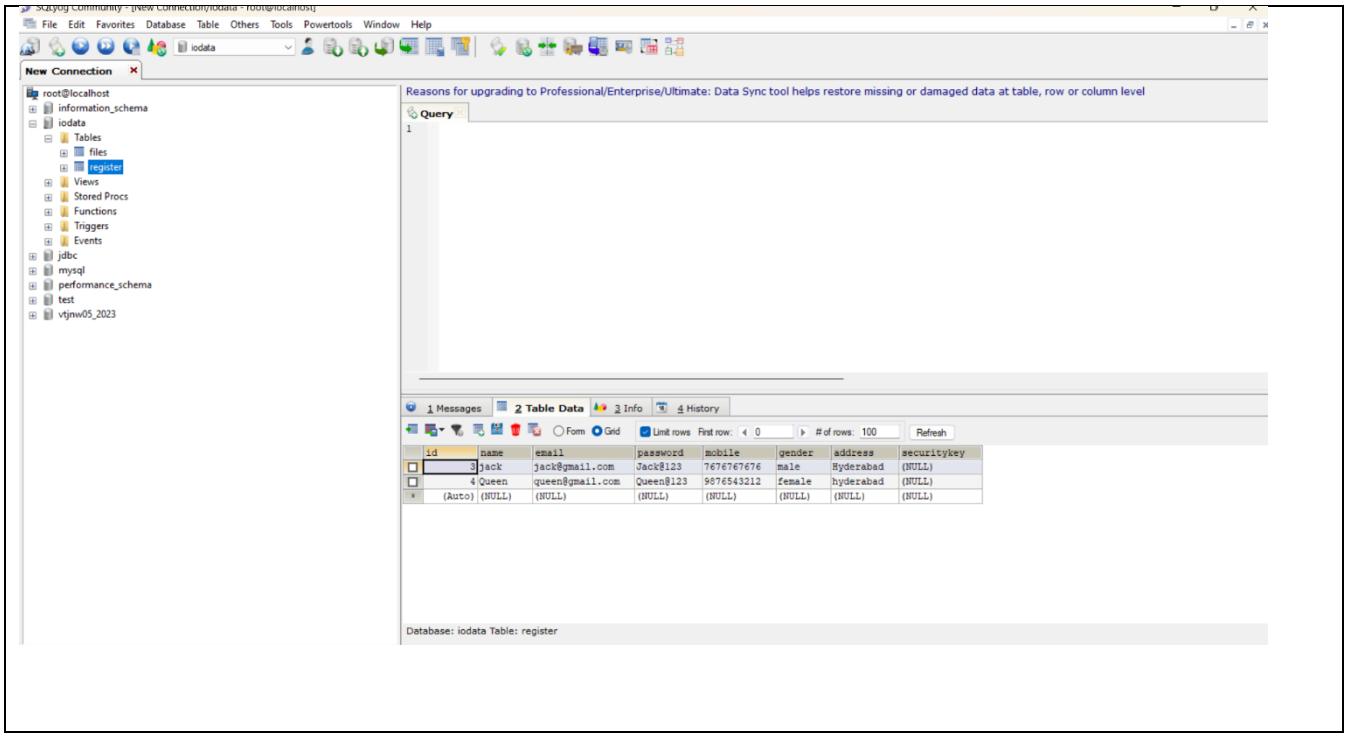


FIGURE 4.2.4: Data Base Page

This shows the register table from the iodata database in SQLyog, storing user details like name, email, password, and gender.

It confirms successful registration by listing stored records, ensuring backend data integrity.

4.2.5 Home Page

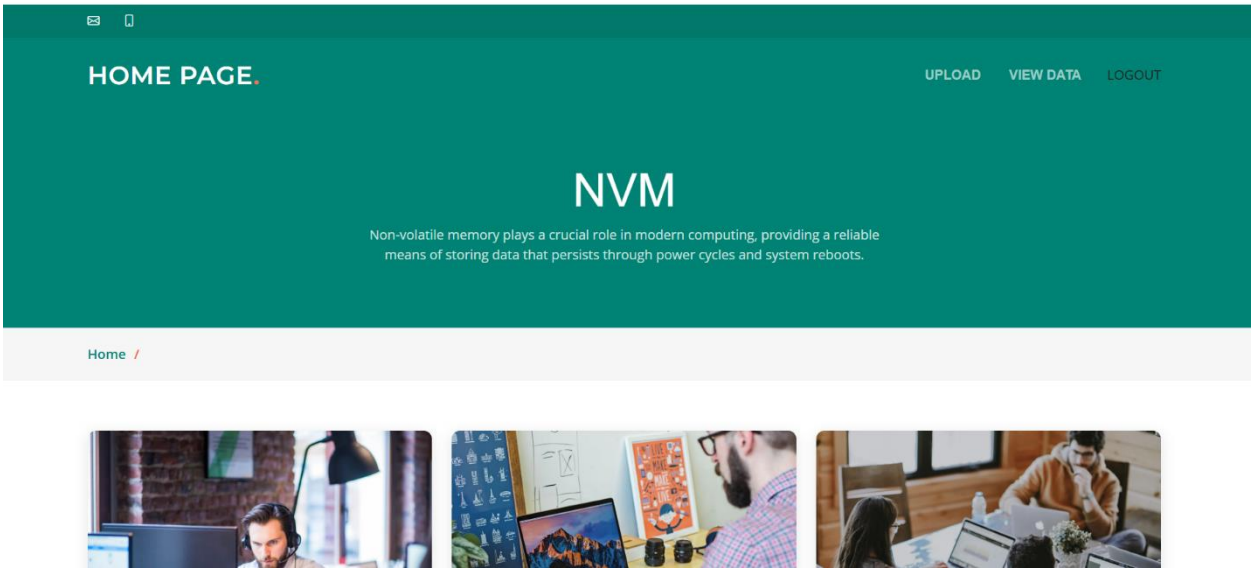


FIGURE 4.2.5: Home Page

The Home Page of the NVM platform provides users with quick navigation options like Upload, View Data, and Logout, while highlighting the significance of Non-Volatile Memory in modern computing.

4.2.6 Upload Page

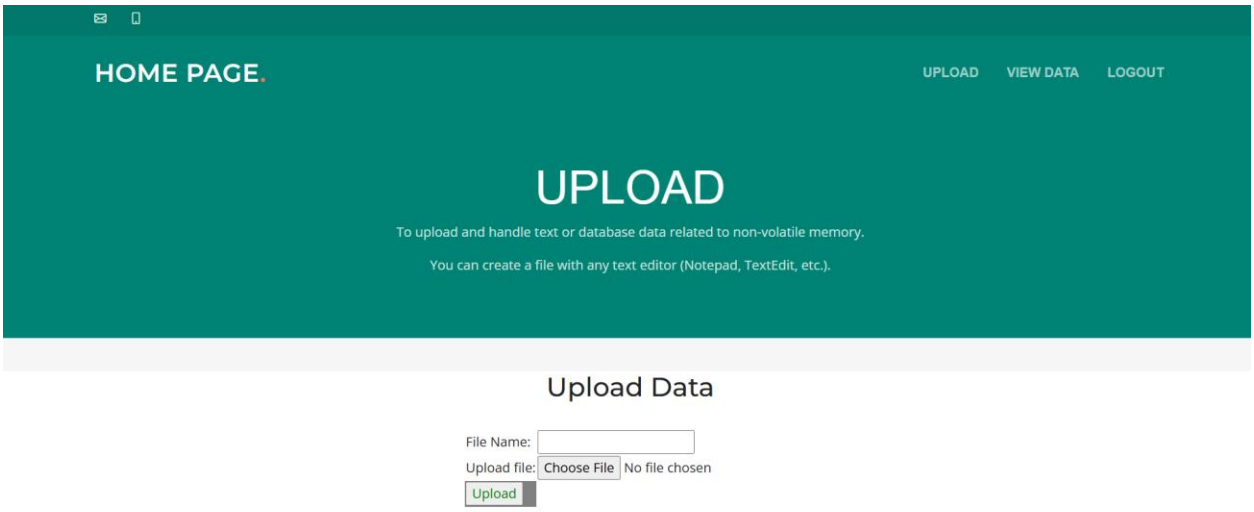


Figure 4.2.6: Upload Page

This page allows users to upload text or database files related to non-volatile memory for processing and storage.

4.2.7 String Comparison Result Page

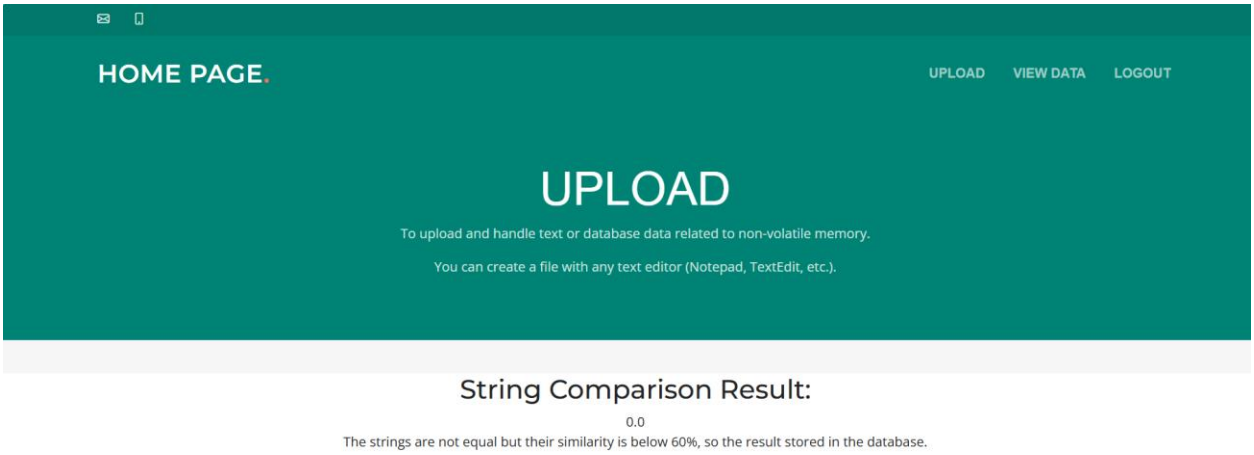


Figure 4.2.7: String Comparison Result Page

This page displays the similarity score between uploaded strings; if the score is below 60%, the data is considered unique and stored in the database.

4.2.8 String Upload Validation Message

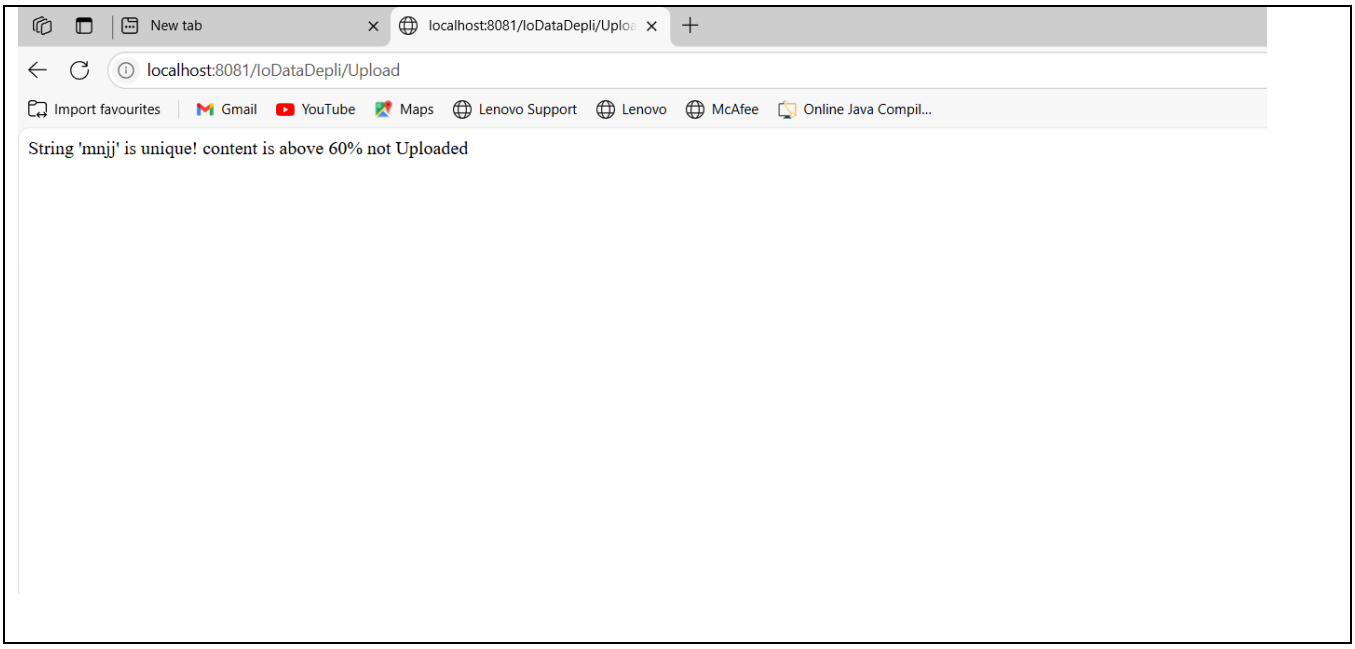


Figure 4.2.8: String Upload Validation Message

This page notifies the user that the uploaded string has over 60% similarity with existing data, so it is not stored in the database to avoid redundancy.

CHAPTER 5

CONCLUSION & FUTURE ENHANCEMENT

5.1 GENERAL

Finally, a rank of data containing the filtered data with correlation value is given as the data feed.

5.2 CONCLUSION

In this project, we present an I/O Causality-based Inline Deduplication (ICID) scheme for NVM-based storage. ICID records the I/O causality among in-memory file operations to achieve in-line data deduplication, and avoid the time-consuming calculation of fingerprints caused by previous chunk-based deduplication schemes. We advocate two key technologies to manage memory-copy operations in a B-tree efficiently, i.e., a hybrid data structure to store memory-copy records, and a location-dependent garbage collection scheme for deleting outdated memory-copy records. Our experimental result demonstrates that ICID achieves up to 16× higher deduplication ratio than state-of-the-art deduplication schemes, and also reduces the time overhead of data deduplication by 47% on average.

5.3 FUTURE ENHANCEMENT

Existing data deduplication systems face severe fragmentation problems. The fragmentation can cause random reads and lowers the throughput of data restoration. Some proposals exploit rewriting schemes to relocate unique chunks, and thus can mitigate the fragmentation problem to some extent. A few proposals exploit other optimizations such as caching to accelerate the data restoration due to fragmentation. However, the fragmentation problem is always accompanied with the data deduplication. We can explore defragmentation technologies to mitigate this problem in an offline manner. On the other hand, since the random read performance of NVMs is much higher than that of SSDs/HDDs the fragmentation is no longer a critical problem for NVM-based storage systems.

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